

Deep Learning for teens

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Batch gradient decent, Stochastic gradient decent vs Mini batch gradient decent

- ▶ Batch gradient decent
 - ▶ We go through **all** the sample in dataset
 - ▶ For large dataset link several millions records that's so hard
- ▶ Stochastic gradient decent
 - ▶ Randomly pick **one** sample(record)
- ▶ Mini batch gradient decent
 - ▶ We use several(batch) samples not all or one per epoch
- ▶ Check the code

Batch gradient decent, Stochastic gradient decent vs Mini batch gradient decent

- ▶ Batch GD $\rightarrow$ one update per epoch (gradient over all samples).
- ▶ SGD $\rightarrow$ many updates per epoch (one per sample).
- ▶ Mini-Batch GD $\rightarrow$ several updates per epoch (one per batch).

Batch gradient decent, Stochastic gradient decent vs Mini batch gradient decent

► Gradient Descent Types in Terms of Epochs

► Batch Gradient Descent

- In **one epoch**, the model sees **all samples at once**.
- It computes the gradient using the **entire dataset**, then makes **one weight update per epoch**.
- ✓ Very stable, ✗ but slow for millions of records.

► Stochastic Gradient Descent (SGD)

- In **one epoch**, the model still sees **all samples**, but **one by one**.
- After each sample, it immediately updates the weights.
- So if you have 1,000 samples, you get **1,000 updates per epoch**.
- ✓ Fast updates, ✗ noisy learning curve.

► Mini-Batch Gradient Descent

- In **one epoch**, the dataset is divided into **batches** (e.g., 32, 64, 128 samples).
- After each batch, the model updates weights.
- Example: dataset = 1,000 samples, batch size = 100 → **10 updates per epoch**.
- ✓ Balanced speed + stability, most common in deep learning.

Dropout layer

- ▶ Dropout(0.2)
 - ▶ randomly turns off 20% of the neurons during training, leaving 80% active.

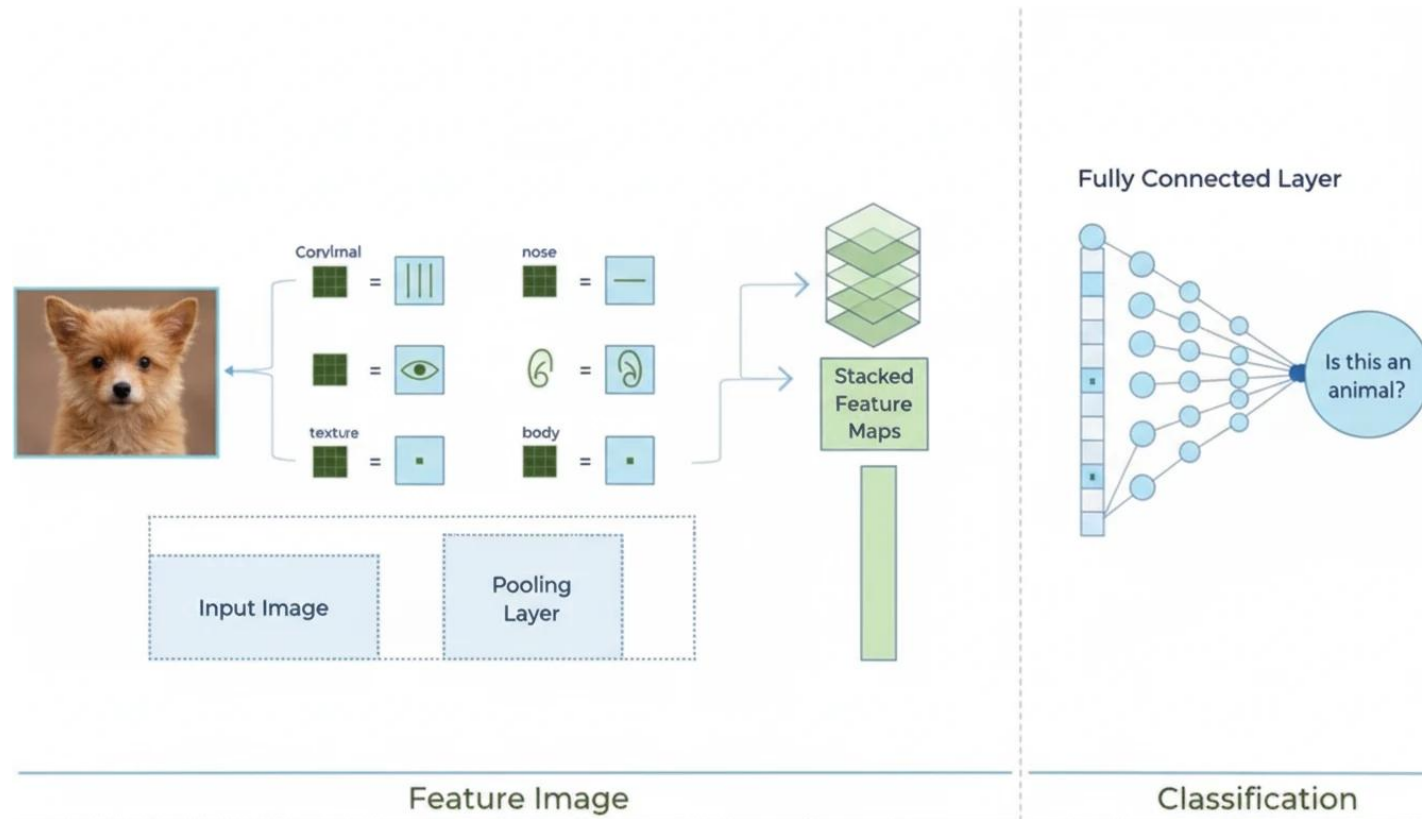
Flatten layer

- ▶ Flatten turns multi-dimensional input into a 1-D vector.

main types of layers in a CNN

- ▶ input Layer: Takes in the raw image data (e.g., 28×28 grayscale pixels).
- ▶ Convolution Layer: Applies filters to detect patterns like edges, textures, or shapes.
- ▶ Activation Layer: Adds non-linearity (usually ReLU) so the network can learn complex features.
- ▶ Pooling Layer: Shrinks the feature maps by keeping only the most important info (e.g., max pooling).
- ▶ Fully Connected Layer: Combines all features and makes the final prediction (e.g., cat vs dog).
- ▶ Output Layer: Gives the final result, often using Softmax for classification.

Convolutional Neural Network (CNN) architecture diagram



ANN vs CNN: Why Convolutions Beat Flattening for Image Classification

- ▶ ANNs flatten the image first → large input layer → many weights → heavy computation.
- ▶ CNNs extract features first → reduce dimensions → fewer weights → better generalization.
- ▶ Pooling helps reduce overfitting and speeds up training.
- ▶ CNNs separate the process into Feature Extraction and Classification, making them more efficient and scalable.

Homework

- ▶ Use the Fashion-MNIST dataset.
- ▶ First, build an ANN.
- ▶ Then, build a CNN.
- ▶ Finally, compare the accuracy of ANN and CNN.